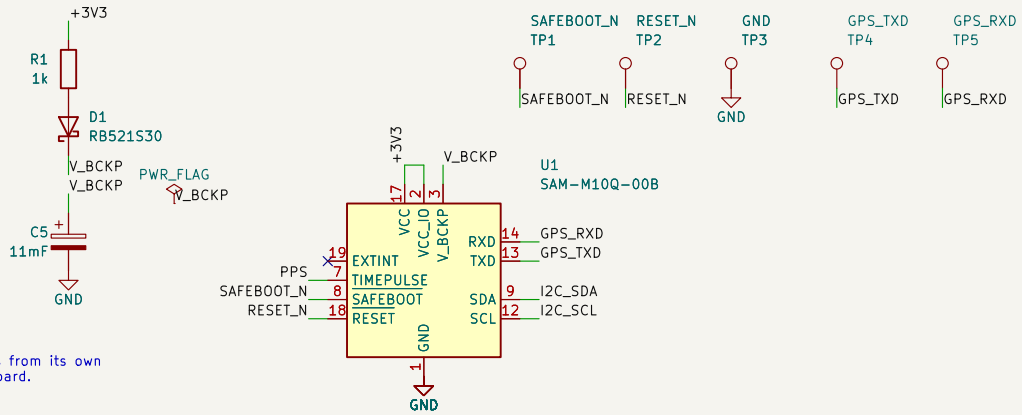
A — Expansion bus (Qwiic / STEMMA QT), 3.3 V from the FC



Qwiic pinout (both ports, wired in parallel):
1 GND
2 +3.3 V (FC expansion LDO, VEXP)
3 SDA
4 SCL
Pull-ups on the FC (R15/R16 2.2 k); R3/R4 here DNP.
Standard 1:1 Qwiic cables.

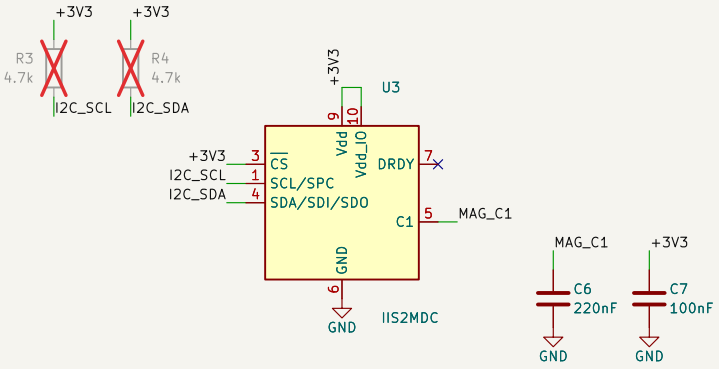
Draft B: no LDO. The FC draft E feeds the expansion bus from its own 3.3 V LDO (U11, VEXP, 300 mA), shared with the flow board.

B — GNSS receiver u-blox SAM-M10Q (antenna module)



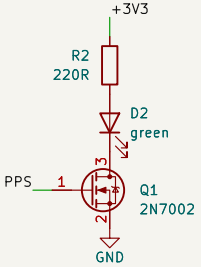
Draft B: the FC reads the GNSS over DDC (I2C, address 0x42) on the Qwiic bus.
UART (TXD/RXD) only on test pads TP4/TP5: u-center / firmware update with a USB-UART.
EXTINT not used. RESET_N: no capacitor (u-blox).
TIMEPULSE shares the pin with the SAFEBOOT strap: only a MOSFET gate on it.

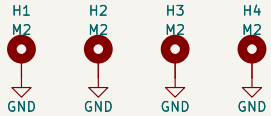
C — Compass IIS2MDC on the Qwiic bus (address 0x1E)



The FC on-board IIS2MDC sits on the LSM6DSV16X sensor-hub bus;
this one is on the Qwiic bus (FC I2C1): GNSS 0x42, ToF 0x29, flow bridge 0x3B: no address clash.
Keep it 5 mm or more from the supercap, the LDO and the cable.

D — PPS indicator, mounting





Stack: FC on its grommets, M2 nylon standoffs (6 mm), GNSS board on top.
Keep-out r 2.6 mm around each hole on both sides.

Generated automatically: run ERC before release
u-blox SAM-M10Q (integrated patch antenna, pin compatible with SAM-M8Q)
Expansion bus: Qwiic (JST SH 4-pin) x 2, pass-through, from FC J3; optional daisy-chain to the flow
30 x 30 mm, 2 layers, 1.0 mm; same 20 x 20 mm hole pattern as the flight controller
ACwing — Andrea Cirillo

Sheet: /
File: ACwing_GPS_A.kicad_sch

Title: ACwing GPS rev A — GNSS + compass expansion for ACwing FC rev A

Size: A4	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1